Trend, Volatility and Market States in E-mini Nasdaq-100 Futures, 1999–2026

Santiago Cruz · Student ID 2584023 · 7CS079 Data Visualisation · University of Wolverhampton

KEY MESSAGE

Twelvefold growth concealed a 16-year loss, clustered storms and fat tails. A 252-day average signals changing risk, not future profits.

1 Introduction

The E-mini Nasdaq-100 futures (NQ) represent one of the most frequently traded derivatives associated with the Nasdaq-100, the index underlying the technology investments held by several retail investors and pension savers (CME Group, n.d.; Nasdaq, 2024). Since 1999, its value has increased twelve times, however long-term averages obscure the irregularity of that expansion. For an investor, the significance lies not alone in the market's growth, but also in the extent, duration, and frequency of its declines along the journey.

Objective: use Tableau to show non-specialist investors and early-career traders what the averages conceal.

O1 Trend RQ1 How deep and how long were the downsides to the trend?
O2 Volatility RQ2 Is risk spread evenly over time and across days, as a bell curve assumes?
O3 Regimes RQ3 Does the 252-day average separate calm from volatile markets?

2 Data source and methodology

Source: daily open, high, low and settlement prices of the continuous front-month contract CME_MINI:NQ1!, 30 Jun 1999 – 30 Oct 2026 (6,914 bars), exported from TradingView (2026), which redistributes CME Globex exchange data.
Why this dataset: 27 years of daily pricing derived from exchanges, covering four crises, pertaining to the tool utilized by specialists to manage Nasdaq-100 risk. ETF valuations incorporate expenses and distributions; standard Kaggle datasets exhibit ambiguous origins. Only public prices are available, therefore eliminating concerns over personal data. Chart selections adhere to the principles established by Munzner (2014) and Tufte (2001): a singular scale for each chart, explicit names, and the color-blind palette provided by Tableau.

1 Audit (Python) — OHLC logic, 0.25-point tick grid and NYSE calendar checked
2 Clean (Python) — 38 holiday sessions and 1 bad print removed: 6,875 sessions
3 Visualise (Tableau) — Measures computed in Python; charts, reference lines and dual axis built in Tableau
4 Test (Statistics) — Welch t, Levene, bootstrap; four averages, two sub-periods

×12.3
price growth since 1999 (9.6% a year)

−83%
worst fall (2000–02); 16 years to regain

117 vs 19
3-sigma days in 27 years: actual vs bell curve

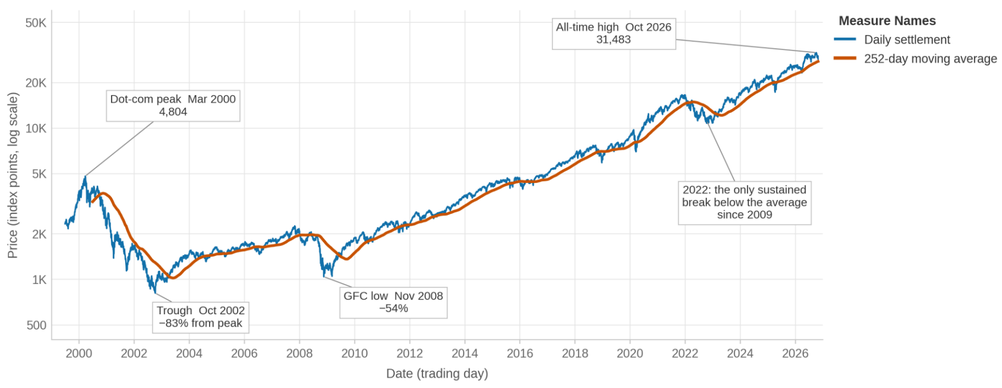
2.3×
volatility below vs above the 252-day average

3 Key visualisations and results

Six Tableau worksheets, tagged by objective: **O1 Trend** **O2 Volatility** **O3 Regimes**

E-mini Nasdaq-100: a 27-year climb from the dot-com peak, through two deep bear markets

Daily settlement of the front-month NQ contract and its 252-day moving average, Jun 1999 – Oct 2026 | ×12.3 (≈9.6% a year)

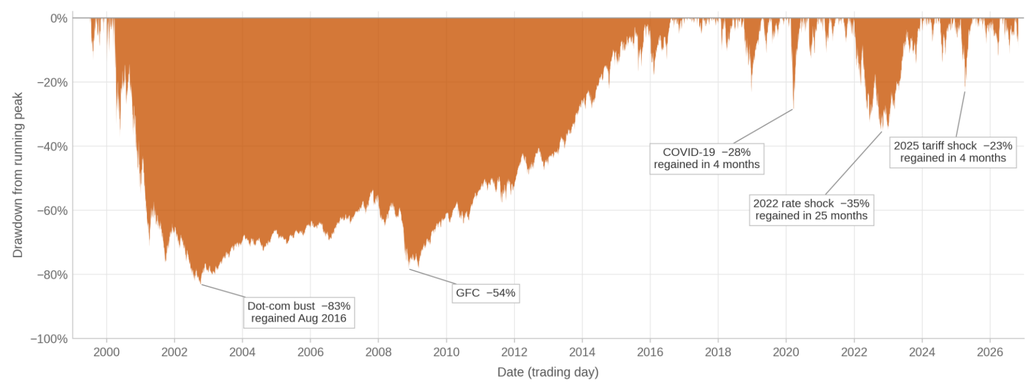


Log scale: equal vertical distances are equal percentage moves. Source: CME Globex via TradingView (CME_MINI:NQ1!).

V1 · O1 Trend Log scale, so equal heights are equal percentage moves. The March 2000 peak of 4,804 fell 83% and was not regained until August 2016. Since 2009 the climb has been steeper, with 2022 the only sustained break below the 252-day average.

Drawdown from previous high: the March 2000 peak took 16 years to regain

Percentage below the highest settlement so far; each label quotes the episode's own peak-to-trough fall

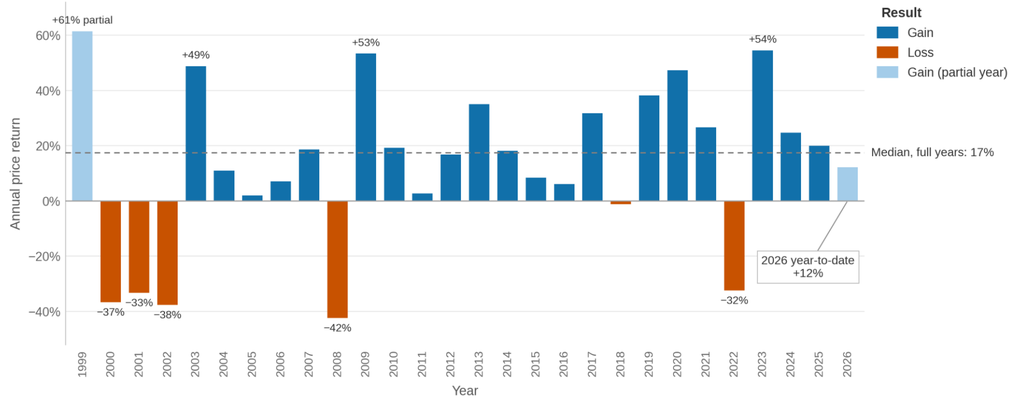


The GFC trough sits inside the dot-com drawdown, so its plotted depth is measured from 2000; its own fall from the Oct 2007 high was −54%.

V2 · O1 Trend Drawdown shows what a holder lived through: the distance below the last peak. The dot-com and GFC losses ran together for 16 years, while COVID (−28%) and the 2025 tariff shock (−23%) recovered within 4 months and 2022 (−35%) in about two years.

Calendar-year return: 20 of 26 full years positive, but losses cluster in runs

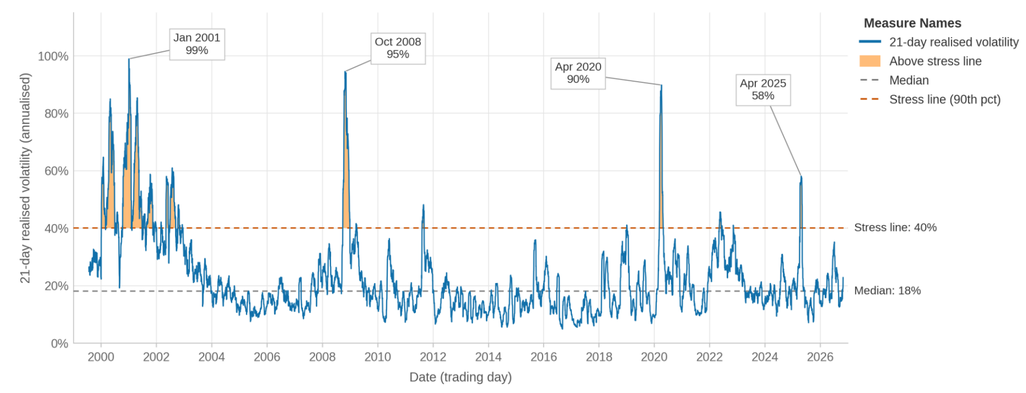
Year-end to year-end change in settlement; 1999 runs from 30 Jun, 2026 is year-to-date (lighter bars)



V3 · O1 Trend 20 of 26 full years rose, with a median gain of 17%, but losses cluster: 2000–02 fell three years running. The two other large losses, 2008 (−42%) and 2022 (−32%), were each followed by a rebound above 50%.

Volatility clusters: calm regimes punctuated by short, violent storms

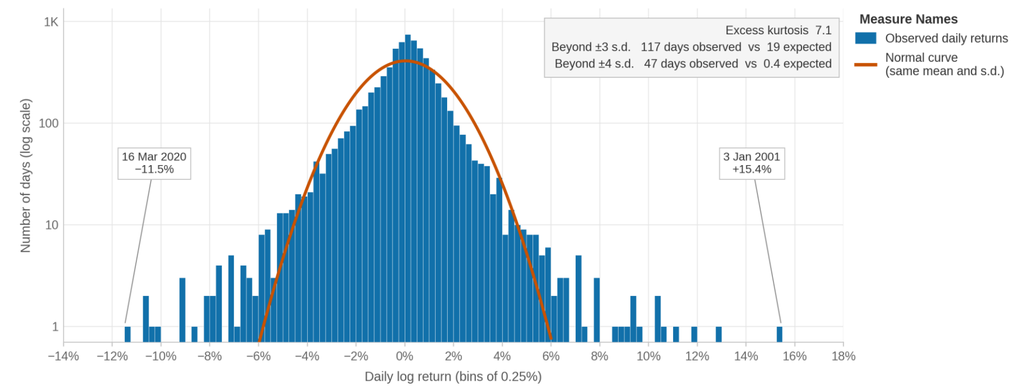
Rolling 21-day standard deviation of daily log returns × √252; shaded where volatility is above the stress line



V4 · O2 Volatility Volatility sits near its 18% median for years, then storms of 58–99% erupt. Storms persist in one, there is a 65% chance of still being in one a month later (base rate 10%). Volatility also rises as prices fall (correlation −0.35).

Daily returns have fat tails: extreme days are far more common than a bell curve predicts

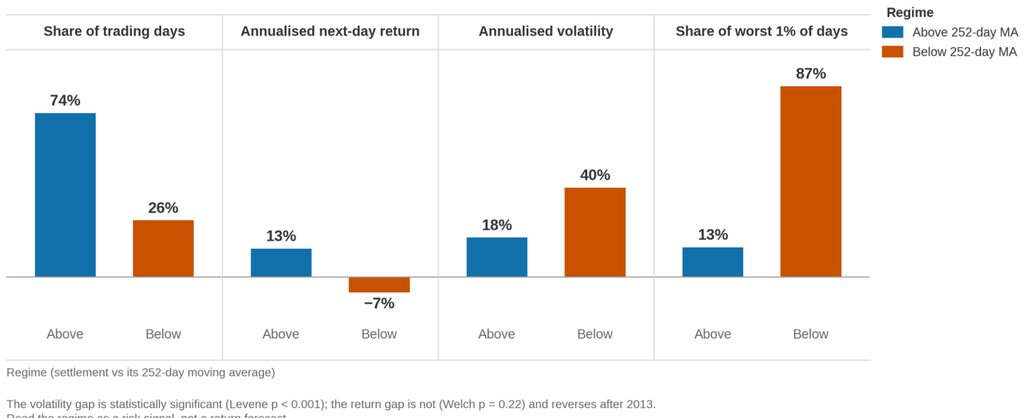
Histogram of 6,872 daily log returns against a normal curve with the same mean and standard deviation



V5 · O2 Volatility A bell curve with the same mean and spread predicts 19 three-sigma days in 27 years; the market produced 117, plus 47 four-sigma days where fewer than one is expected. Excess kurtosis is 7.1 and normality is rejected (Jarque-Bera p < 0.001).

Trend regimes: below its 252-day average, risk more than doubles and most crash days occur

Regime at today's settlement paired with the next day's return (no look-ahead), Jun 2000 – Oct 2026



The volatility gap is statistically significant (Levene p < 0.001); the return gap is not (Welch p = 0.22) and reverses after 2013.
Read the regime as a risk signal, not a return forecast.

V6 · O3 Regimes Below its 252-day average the contract spent only 26% of days, yet volatility more than doubled (40% vs 18%) and 87% of the worst 1% of days occurred. The return gap is not significant (Welch p = 0.22) and reverses after 2013.

4 Discussion and interpretation

O1 Trend Entry date mattered as much as holding period: a buyer at the March 2000 peak (4,804) saw 809.5 by October 2002 and broke even only in August 2016; a buyer at the first 2009 settlement (1,253) held ×22.8 by October 2026 (28,534).
O2 Volatility Clustering and fat tails reproduce the stylized facts of returns (Mandelbrot, 1963; Cont, 2001). They are not an artefact of noisy early data: after 2003, excess kurtosis is 8.1 and 3-sigma days number 83 against 16 expected.
O3 Regimes The 252-day line separates risk for every average length (50–252 days: 36–41% vs 18–20% volatility) and in both sub-periods, but not returns (Welch p = 0.22–0.75), and the price crossed it 140 times with a median spell of five days. This fits the Adaptive Markets view (Lo, 2004) better than a trading rule.
So what? Non-specialists should judge an investment by its drawdowns and storms, not only its average return. These in-sample results for one contract describe past risk; they do not forecast it.

5 Conclusion

NQ's long-run growth (×12.3, from 2,325.50 to 28,534) was real but unrepresentative: it came with deep, clustered and fat-tailed risk.

RQ1: deep and uneven: −83.1% and 16.4 years to recover, with the GFC's −53.9% inside it; then −28.4% (2020), −35.3% (2022) and −22.5% (2025), recovered in 3.5, 24.7 and 4.1 months.

RQ2: no. Storms persist (65% vs a 10% base rate); 117 three-sigma days vs 19 expected.

RQ3: yes for risk (40.5% vs 17.9% volatility), no for returns (p = 0.22).

Recommendations: show drawdown and volatility on dashboards; treat the 252-day line as a risk thermometer.

Limitations: bubble-peak start, unadjusted rolls; explore GARCH and regime-switching models next.

References

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